Algebraic Methods

Algebraic Methods

Solution 2

1(a) Simplify $3x^2 \times 5x^5$.

1(a) $15x^7$

1(b) Simplify $\frac{5x^3y^2}{15x^2y^3}$.

1(b) $\frac{x}{3y}$

2(a) Factorise $x^2 - 2x - 24$.

2(a) $(x-6)(x+4)$

2(b) Factorise $3x^2 - 17x + 20$.

2(b) $(3x-5)(x-4)$

3(a) Calculate $197041 \div 23$ using long division.

3(a) 8567

3(b) Calculate $56168 \div 34$ using long division.

3(b) 1652

4(a) Find the line through $(-1,4)$ and $(5,-14)$.

4(a) $y = 1 - 3x$

4(b) Find the line through $(2,-6)$ and $(8,-3)$.

4(b) $y = \frac{1}{2}x - 7$

5(a) Complete the square: $x^2 - 2x - 20$.

5(a) $(x-1)^2 - 21$

5(b) Complete the square: $2x^2 + 4x + 15$.

5(b) $2(x+1)^2 + 13$

Chapter Overview

Solution 3

1. Algebraic fractions

Factorise and cancel common factors.

2. Dividing polynomials

Use long division, including remainders.

3. The factor theorem

Identify factors and determine unknown coefficients.

4. Proof

Deduction, exhaustion and counterexamples.

1: Simplifying Algebraic Fractions

Solution 4

Factorise first, then cancel common factors. Never cancel terms joined by addition or subtraction. Retain the restrictions from the original denominator.

1

$$\frac{x^2-1}{x^2+x}$$

1

$$\frac{(x+1)(x-1)}{x(x+1)} = \frac{x-1}{x}$$

2

$$\frac{x^2+3x+2}{x+1}$$

2

$$\frac{(x+2)(x+1)}{x+1} = x+2$$

3

$$\frac{2x^2+11x+12}{x^2+9x+20}$$

3

$$\frac{(2x+3)(x+4)}{(x+4)(x+5)} = \frac{2x+3}{x+5}$$

4

$$\frac{4-x^2}{x^2+2x-8}$$

4

$$\frac{-(x-2)(x+2)}{(x+4)(x-2)} = -\frac{x+2}{x+4}$$

Tips: factorise the easier expression first. Also $\frac{a-b}{b-a} = -1$. Do not leave 1 as a denominator.

Exercise 7A – Y1 Pages 138-139 Solution 5

2. Simplify as far as possible.

m

$$\frac{2x^2+3x+1}{x^2-x-2}$$

m

$$\frac{(2x+1)(x+1)}{(x-2)(x+1)} = \frac{2x+1}{x-2}$$

n

$$\frac{x^2+6x+8}{3x^2+7x+2}$$

n

$$\frac{(x+4)(x+2)}{(3x+1)(x+2)} = \frac{x+4}{3x+1}$$

o

$$\frac{2x^2-5x-3}{2x^2-9x+9}$$

o

$$\frac{(2x+1)(x-3)}{(2x-3)(x-3)} = \frac{2x+1}{2x-3}$$

3 Find a,b,c given

$$\frac{6x^3+3x^2-84x}{6x^2-33x+42} = \frac{ax(x+b)}{x+c}$$

3

$$\frac{3x(2x-7)(x+4)}{3(x-2)(2x-7)} = \frac{x(x+4)}{x-2}$$

$$a=1, \quad b=4, \quad c=-2$$

Terminology Solution 6

$11 \div 4 = 2$ remainder 3

Dividend The number being divided

Dividend

11

Divisor The number we divide by

Divisor

4

Quotient The whole-number result

Quotient

2

Remainder The amount left over

Remainder

3

Dividend = divisor × quotient + remainder

Normal Long Division

Solution 7

Calculate $423 \div 11$ using long division.

1 Divide 42 by 11.

1

3 whole times. Write 3 above the tens column.

$42 - 3 \times 11 = 9$

2 Bring down the final digit, 3.

2

Now divide 93 by 11: 8 whole times.

$93 - 8 \times 11 = 5$

3 State the quotient and remainder.

3

$423 = 11 \times 38 + 5$

$423 \div 11 = 38$ remainder 5

4 Continue to a decimal.

4

$50 - 4 \times 11 = 6$

$60 - 5 \times 11 = 5$

$423 \div 11 = 38.4545\ldots$

Repeat: divide, multiply by the divisor, subtract, bring down the next digit.

Example

Solution 8

Divide

$$\frac{6x^3 + 28x^2 - 7x + 15}{x + 5}$$

1

$6x^3 \div x = 6x^2$

1

$(6x^3 + 28x^2) - (6x^3 + 30x^2) = -2x^2$

2

$-2x^2 \div x = -2x$

2

$(-2x^2 - 7x) - (-2x^2 - 10x) = 3x$

3

$3x \div x = 3$

3

$(3x + 15) - (3x + 15) = 0$

Result Combine the quotient terms and remainder.

Result

$6x^2 - 2x + 3$

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

Further Example

Solution 9

Divide

$$\frac{3x^3 + 0x^2 - 2x + 4}{x - 1}$$

1

$3x^3 \div x = 3x^2$

1

$$(3x^3 + 0x^2) - (3x^3 - 3x^2) = 3x^2$$

2

$3x^2 \div x = 3x$

2

$$(3x^2 - 2x) - (3x^2 - 3x) = x$$

3

$x \div x = 1$

3

$$(x + 4) - (x - 1) = 5$$

Result Combine the quotient terms and remainder.

Result

$$3x^2 + 3x + 1 + \frac{5}{x - 1}$$

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

Test Your Understanding

Solution 10

Divide

$$\frac{2x^3 - 5x^2 - 16x + 10}{x - 4}$$

1

$2x^3 \div x = 2x^2$

1

$$(2x^3 - 5x^2) - (2x^3 - 8x^2) = 3x^2$$

2

$3x^2 \div x = 3x$

2

$$(3x^2 - 16x) - (3x^2 - 12x) = -4x$$

3

$-4x \div x = -4$

3

$$(-4x + 10) - (-4x + 16) = -6$$

Result Combine the quotient terms and remainder.

Result

$$2x^2 + 3x - 4 - \frac{6}{x - 4}$$

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

Further Test Your Understanding — Solution 11

Divide

$$\frac{8x^3 + 0x^2 + 0x - 1}{2x - 1}$$

1

$8x^3 \div 2x = 4x^2$

1

$(8x^3 + 0x^2) - (8x^3 - 4x^2) = 4x^2$

2

$4x^2 \div 2x = 2x$

2

$(4x^2 + 0x) - (4x^2 - 2x) = 2x$

3

$2x \div 2x = 1$

3

$(2x - 1) - (2x - 1) = 0$

Result Combine the quotient terms and remainder.

Result

$4x^2 + 2x + 1$

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

Extension: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

Exercise 7B – Y1 Pages 142 — Solution 12

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.

Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.

a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.

b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

15

$$\begin{array}{r|l} & 2x^2 + 7x - 6 \\ 5x + 4 & 10x^3 + 43x^2 - 2x - 10 \\ & \underline{10x^3 + 8x^2} \\ & \quad 35x^2 - 2x \\ & \quad \underline{35x^2 + 28x} \\ & \quad\quad -30x - 10 \\ & \quad\quad \underline{-30x - 24} \\ & \quad\quad\quad 14 \end{array}$$

So the remainder is 14.

16(a)

a

$$\begin{array}{r|l} & 3x^2 - 5x - 62 \\ x - 3 & 3x^3 - 14x^2 - 47x - 14 \\ & \underline{3x^3 - 9x^2} \\ & \quad -5x^2 - 47x \\ & \quad \underline{-5x^2 + 15x} \\ & \quad\quad -62x - 14 \\ & \quad\quad \underline{-62x + 186} \\ & \quad\quad\quad -200 \end{array}$$

So the remainder is -200.

16(b)

b

$$\begin{array}{r|l} & 3x^2 - 20x - 7 \\ x + 2 & 3x^3 - 14x^2 - 47x - 14 \\ & \underline{3x^3 + 6x^2} \\ & \quad -20x^2 - 47x \\ & \quad \underline{-20x^2 - 40x} \\ & \quad\quad -7x - 14 \\ & \quad\quad \underline{-7x - 14} \\ & \quad\quad\quad 0 \end{array}$$

$$\begin{aligned} \mathrm{f}(x) &= 3x^3 - 14x^2 - 47x - 14 \\ &= (x + 2)(3x^2 - 20x - 7) \\ &= (x + 2)(3x + 1)(x - 7) \end{aligned}$$

Exercise 7B – Y1 Pages 142 — Solution 13

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by

i $x - 2$,

ii $x + 3$.

b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

a(i)

i

$$\begin{array}{r|l} & x^2 + 8x + 21 \\ x - 2 & x^3 + 6x^2 + 5x - 12 \\ & \underline{x^3 - 2x^2} \\ & \quad 8x^2 + 5x \\ & \quad \underline{8x^2 - 16x} \\ & \quad\quad 21x - 12 \\ & \quad\quad \underline{21x - 42} \\ & \quad\quad\quad 30 \end{array}$$

So the remainder is 30.

a(ii)

ii

$$\begin{array}{r|l} & x^2 + 3x - 4 \\ x + 3 & x^3 + 6x^2 + 5x - 12 \\ & \underline{x^3 + 3x^2} \\ & \quad 3x^2 + 5x \\ & \quad \underline{3x^2 + 9x} \\ & \quad\quad -4x - 12 \\ & \quad\quad \underline{-4x - 12} \\ & \quad\quad\quad 0 \end{array}$$

So the remainder is 0.

b

b

$$\begin{aligned} f(x) &= x^3 + 6x^2 + 5x - 12 \\ &= (x + 3)(x^2 + 3x - 4) \\ &= (x + 3)(x + 4)(x - 1) \end{aligned}$$

So $x = -3, x = -4$ or $x = 1$.

Exercise 7B – Y1 Pages 142 — Solution 14

18 $f(x) = 2x^3 + 3x^2 - 8x + 3$.

a Show that $f(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.

b Hence factorise $f(x)$ completely.

c Write down all the real roots of the equation $f(x) = 0$.

a

a

$$\begin{array}{r|l} & x^2 + 2x - 3 \\ 2x - 1 & 2x^3 + 3x^2 - 8x + 3 \\ & \underline{2x^3 - x^2} \\ & \quad 4x^2 - 8x \\ & \quad \underline{4x^2 - 2x} \\ & \quad\quad -6x + 3 \\ & \quad\quad \underline{-6x + 3} \\ & \quad\quad\quad 0 \end{array}$$

$$\begin{aligned} f(x) &= 2x^3 + 3x^2 - 8x + 3 \\ &= (2x - 1)(x^2 + 2x - 3) \end{aligned}$$

$a = 1, b = 2, c = -3$.

b

b

$$\begin{aligned} f(x) &= 2x^3 + 3x^2 - 8x + 3 \\ &= (2x - 1)(x^2 + 2x - 3) \\ &= (2x - 1)(x - 1)(x + 3) \end{aligned}$$

c

c $(2x - 1)(x - 1)(x + 3) = 0$.

$x = \frac{1}{2}, x = 1$ and $x = -3$.

Exercise 7B – Y1 Pages 142 Solution 15

19 f$(x) = 12x^3 + 5x^2 + 2x - 1$.

a Show that $(4x - 1)$ is a factor of f(x) and write f(x) in the form $(4x - 1)(ax^2 + bx + c)$.

b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

a

a

$$
\begin{array}{r|l}
 & 3x^2 + 2x + 1 \\
4x - 1 & 12x^3 + 5x^2 + 2x - 1 \\
 & \underline{12x^3 - 3x^2} \\
 & \quad 8x^2 + 2x \\
 & \quad \underline{8x^2 - 2x} \\
 & \quad\quad 4x - 1 \\
 & \quad\quad \underline{4x - 1} \\
 & \quad\quad\quad 0
\end{array}
$$

f$(x) = (4x - 1)(3x^2 + 2x + 1)$.

$a = 3, b = 2, c = 1$.

b

b $(4x - 1)(3x^2 + 2x + 1) = 0$.

Using the discriminant for $3x^2 + 2x + 1$:

$b^2 - 4ac = 2^2 - 4(3)(1) = -8 < 0$ so there are no real solutions.

So f(x) has only one real solution, $x = \frac{1}{4}$.

3: The Factor Theorem Solution 16

Factor theorem

$(x - p)$ is a factor of a polynomial f(x) if and only if f(p)=0.

Example For $f(x) = x^3 + x^2 - 4x - 4$, test whether x−2 is a factor.

Example

$f(2) = 8 + 4 - 8 - 4 = 0$

Therefore x−2 is a factor.

$f(x) = (x - 2)(x^2 + 3x + 2)$

Examples (in textbook) Solution 17

1 Show that x−2 is a factor of $x^3 + x^2 - 4x - 4$.

1

$f(2) = 0$

2: test Factorise $f(x) = 2x^3 + x^2 - 18x - 9$ completely. First find one root.

2: test

$f(1) = -24, \quad f(-1) = 8, \quad f(3) = 0$

Thus x−3 is a factor.

2: divide Find the quadratic factor.

2: divide

$f(x) = (x-3)(2x^2 + 7x + 3)$

2: factorise Factorise the quadratic.

2: factorise

$f(x) = (x-3)(2x+1)(x+3)$

Alternatively compare coefficients: in $(x-3)(ax^2+bx+c)$, a=2, c=3 and b−3a=1 gives b=7. Try integer divisors of the constant term, or use a calculator table, to find a root.

Using Factor Theorem to find unknown coefficients Solution 18

Given that 2x+1 is a factor of $6x^3 + ax^2 + 1$, find a.

1 Choose the value making the factor zero.

1

$2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$

2 Apply the factor theorem.

2

$6(-\frac{1}{2})^3 + a(-\frac{1}{2})^2 + 1 = 0$

$-\frac{3}{4} + \frac{a}{4} + 1 = 0 \Rightarrow a = -1$

For a linear factor ax+b, substitute x=−b/a into the polynomial.

Test Your Understanding — Solution 19

$f(x) = 6x^3 + 13x^2 - 4$

1(a) Find the remainder when divided by 2x+3.

1(a)

$$f(-\frac{3}{2}) = -\frac{81}{4} + \frac{117}{4} - 4 = 5$$

1(b) Show that x+2 is a factor.

1(b)

$$f(-2) = -48 + 52 - 4 = 0$$

1(c) Factorise f(x) completely.

1(c)

$$f(x) = (x+2)(6x^2 + x - 2)$$
$$= (x+2)(2x-1)(3x+2)$$

2 Given that 3x−1 is a factor of $3x^3 + 11x^2 + ax + 1$, find a.

2

$$f(\frac{1}{3}) = \frac{1}{9} + \frac{11}{9} + \frac{a}{3} + 1 = 0$$
$$\frac{7}{3} + \frac{a}{3} = 0 \Rightarrow a = -7$$

Exercise 7C – Y1 Pages 146 — Solution 20

13. $f(x) = 3x^3 - 12x^2 + 6x - 24$.

a Use the factor theorem to show x−4 is a factor.

a

$$f(4) = 192 - 192 + 24 - 24 = 0$$

Therefore x−4 is a factor.

b Show that 4 is the only real root.

b

$$f(x) = (x-4)(3x^2 + 6)$$

Polynomial division gives quotient $3x^2+6$ and remainder zero.

$3x^2 + 6 > 0$ for every real x

Thus the only real root is x=4.

Exercise 7C – Y1 Pages 145-146 Solution 21

14. $f(x) = 4x^3 + 4x^2 - 11x - 6$.

a Show x+2 is a factor.

a

$f(-2) = -32 + 16 + 22 - 6 = 0$

Therefore x+2 is a factor.

b Factorise f(x) completely.

b

$$\begin{array}{r|l} & 4x^2 - 4x - 3 \\ x + 2 & 4x^3 + 4x^2 - 11x - 6 \\ & \underline{4x^3 + 8x^2} \\ & \quad -4x^2 - 11x \\ & \quad \underline{-4x^2 - 8x} \\ & \quad\quad -3x - 6 \\ & \quad\quad \underline{-3x - 6} \\ & \quad\quad\quad 0 \end{array}$$

$f(x) = (x + 2)(4x^2 - 4x - 3)$

$= (x + 2)(2x - 3)(2x + 1)$

c Solve f(x)=0.

c

$x = -2, \quad x = \frac{3}{2}, \quad x = -\frac{1}{2}$

Exercise 7C – Y1 Pages 145-146 Solution 22

15. $f(x) = 9x^4 - 18x^3 - x^2 + 2x$.

a Show that x−2 is a factor.

a

$f(2) = 144 - 144 - 4 + 4 = 0$

Therefore x−2 is a factor.

b Find all four real roots.

b

$f(x) = (x - 2)(9x^3 - x)$

$= x(x - 2)(9x^2 - 1)$

$= x(x - 2)(3x + 1)(3x - 1)$

$x = 0, \quad 2, \quad -\frac{1}{3}, \quad \frac{1}{3}$

Exercise Extension

Solution 23

1 [MAT 2006] $x^3 + ax + b$ has factors x−1 and x−2. Find a,b.

1

$1 + a + b = 0, \quad 8 + 2a + b = 0$

$a = -7, \quad b = 6$

2 [MAT 2009] For which positive integers n does $x^2 - 1$ divide $n^2x^{2n+3} - 25nx^{n+1} + 150x^7$?

2

$f(1) = (n - 10)(n - 15) = 0$

Test n=10 and 15 at x=−1. For even n, f(−1)=−f(1)=0; for odd n, f(−1)=−n²−25n−150≠0. Only n=10 works.

3 [MAT 2013] $p_n(x) = (x - 1) + \cdots + (x - n)$, n≥2. Find the remainder on division by $p_{n-1}(x)$.

3

$p_n(x) = nx - \frac{n(n+1)}{2}$

$p_{n-1}(x) = (n - 1)(x - \frac{n}{2})$

Remainder $= p_n(\frac{n}{2}) = -\frac{n}{2}$

Remainder theorem: division by x−a leaves remainder f(a). For any nonconstant linear divisor, evaluate f at its root.

4: Proof

Solution 24

Conjecture

A mathematical statement not yet proved.

Goldbach's conjecture: every even integer greater than 2 is a sum of two primes. Checking many cases supplies evidence, but does not prove every case.

Theorem

A mathematical statement that has been proved.

Fermat's Last Theorem: for integers n>2 there are no positive integers a,b,c satisfying $a^n + b^n = c^n$. Its proof was published in 1995.

Types of Proof

Solution 25

A proof states its assumptions, follows a logical sequence, covers every case and ends with a conclusion. Deduction starts with known facts.

1 Prove that the product of two odd integers is odd.

1

Let p,q be integers. The odd integers have forms 2p+1 and 2q+1.

$(2p+1)(2q+1) = 4pq + 2p + 2q + 1$

$= 2(2pq + p + q) + 1$

This is one more than an even integer, so the product is odd.

2 Prove $(3x+2)(x-5)(x+7) \equiv 3x^3 + 8x^2 - 101x - 70$.

2

$(3x+2)(x^2+2x-35)$

$= 3x^3 + 6x^2 - 105x + 2x^2 + 4x - 70$

$= 3x^3 + 8x^2 - 101x - 70$

An identity holds for every permissible value: $x(x-1) \equiv x^2 - x$. By contrast, $x^2 = 4$ holds only for x=±2.

Be Warned...

Solution 26

Prove that consecutive integer side lengths of a right-angled triangle must be 3,4,5.

Insufficient argument

Checking $3^2+4^2=5^2$ shows one example works. It does not exclude other consecutive triples.

Proof

Let the positive integer sides be x,x+1,x+2. By Pythagoras,

$x^2 + (x+1)^2 = (x+2)^2$

$x^2 - 2x - 3 = (x-3)(x+1) = 0$

Since x>0, x=3. Therefore the sides must be 3,4,5.

Start from the given assumptions, rather than assuming the conclusion you are asked to prove.

Types of Proof

Solution 27

Example Prove $x^2 + 4x + 5$ is positive for all real x.

Example

$x^2 + 4x + 5 = (x + 2)^2 + 1 \geq 1 > 0$

Your turn Prove that the sum of the squares of consecutive odd integers is 2 more than a multiple of 8.

Your turn

Write them as 2n−1 and 2n+1, for integer n.

$(2n - 1)^2 + (2n + 1)^2$

$= 4n^2 - 4n + 1 + 4n^2 + 4n + 1$

$= 8n^2 + 2$

This is 2 more than a multiple of 8.

Exercise 7D – Y1 Pages 149-150

Solution 28

11 Prove that $x^2 + 8x + 20 \geq 4$ for all values of x.

11

The minimum value of $(x + 4)^2 + 4$ is 4.

So $(x + 4)^2 + 4 \geq 4$.

Therefore, $x^2 + 8x + 20 \geq 4$.

12 The equation $kx^2 + 5kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < \frac{12}{25}$.

12

$kx^2 + 5kx + 3 = 0$ has no real roots, so

$b^2 - 4ac < 0$.

$(5k)^2 - 4k(3) < 0$

$25k^2 - 12k < 0$

$k(25k - 12) < 0$

$0 < k < \frac{12}{25}$

When $k = 0$:

$(0)x^2 + 5(0)x + 3 = 0, \qquad 3 = 0$

which is impossible, so no real roots.

So combining these:

$0 \leq k < \frac{12}{25}$

Exercise 7D – Y1 Pages 149-150 Solution 29

13 $px^2 - 5x - 6 = 0$ has two distinct real roots. Prove $p > -\frac{25}{24}$.

13

$D = 25 + 24p > 0$

$p > -\frac{25}{24}$

Two roots also imply p≠0.

18 $y = ax$ cuts $(x-1)^2 + y^2 = k$, k>0, at two distinct points. Prove $k > \frac{a^2}{1+a^2}$.

18

$(1+a^2)x^2 - 2x + 1 - k = 0$

The discriminant is positive:

$4 - 4(1+a^2)(1-k) > 0$

$k(1+a^2) > a^2$

Divide by $1+a^2>0$ to obtain the result.

Exercise 7D – Y1 Pages 149-150 Solution 30

19 Prove that $4y - 3x + 26 = 0$ is tangent to $(x+4)^2 + (y-3)^2 = 100$.

19

$y = \frac{3}{4}x - \frac{13}{2}$

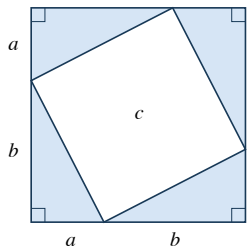
$(x+4)^2 + (\frac{3}{4}x - \frac{19}{2})^2 = 100$

$25x^2 - 100x + 100 = 0$

$25(x-2)^2 = 0$

One repeated intersection, at $(2,-5)$, so the line is tangent.

20. The diagram shows a square and four congruent right-angled triangles. Use areas to prove $a^2 + b^2 = c^2$.



Area proof

Outer area $= (a+b)^2$

Four triangles $= 4 \times \frac{1}{2}ab = 2ab$

$c^2 = (a+b)^2 - 2ab = a^2 + b^2$

Exercise Extension

Solution 31

[STEP I 2005 Q1] Prove there are 15 five-digit numbers with digit sum 43. How many have digit sum 39? Explain your reasoning.

i Digit sum 43

i

The maximum is 45 from 99999. A deficit of 2 gives digits (9,9,9,9,7), with 5 arrangements, or (9,9,9,8,8), with 10 arrangements.

$5 + 10 = 15$

ii Digit sum 39

ii

A deficit of 6 gives these digit sets and numbers of arrangements:

99993	5
99984	20
99975	20
99885	30
99966	10
99876	60
98886	20
99777	10
98877	30
88887	5

Total = 210

Other Types of Proof

Solution 32

Proof by exhaustion

Divide the possibilities into cases and prove each case.

1 Prove $n^2 + n$ is even for every integer n.

1

If n is even, n^2+n is even+even, hence even. If n is odd, n^2+n is odd+odd, hence even. These cover every integer, so the claim holds.

Disproof by counterexample

One counterexample disproves a statement claiming every case works.

2 Disprove: $n^2 - n + 41$ is prime for every integer n.

2

$n = 41 \Rightarrow n^2 - n + 41 = 41^2$

This is composite, so the claim is false.

Exercise 7E – Y1 Pages 152-154 Solution 33

4 Prove each integer cube is a multiple of 9, or one more or one less.

4

Every integer is 3n,3n+1 or 3n+2.

$(3n)^3 = 9(3n^3)$

$(3n+1)^3 = 9(3n^3 + 3n^2 + n) + 1$

$(3n+2)^3 = 9(3n^3 + 6n^2 + 4n + 1) - 1$

The three cases give the required remainders.

7 Prove $(x+6)^2 \geq 2x + 11$ for every real x.

7

$(x+6)^2 - (2x+11) = (x+5)^2 \geq 0$

Therefore the stated inequality holds.

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8 For a>0, prove $a + \frac{1}{a} \geq 2$.

8

$(a-1)^2 \geq 0 \Rightarrow a^2 + 1 \geq 2a$

Divide by positive a:

$$a + \frac{1}{a} \geq 2$$

9(a) For p,q>0, prove $p + q \geq \sqrt{4pq}$.

9(a)

$(p+q)^2 = (p-q)^2 + 4pq \geq 4pq$

Since p+q>0, taking square roots gives the result.

Equality occurs when p=q.

9(b) Show by counterexample that the inequality need not hold when both numbers are negative.

9(b)

$p = q = -1: \quad p + q = -2, \quad \sqrt{4pq} = 2$

But $-2<2$, so the inequality fails.